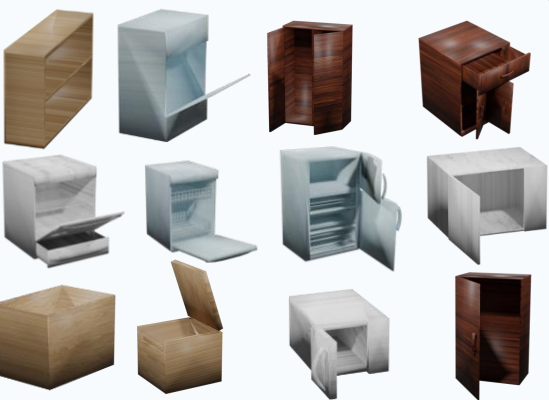


(a) Scene Context



- Robot Specifications
- Table Specifications
- Task / Scene Description

(b) Asset Pool



(c) LLM-Guided Workspace Synthesis

LLM Prompt #1

Primitive Type / Count
Selection

Select the primitive types and quantities for each table based on the robot, table geometry, primitive pool, and scene description.



LLM [1]



Table 1: Cabinet, Open Box, Microwave.

Table 2: Open Box.

Table 3: Shelf

Table 4: Cabinet

Procedural Primitive Generation

Instantiate Selected
Primitives



Mesh models with randomized dimensions, and articulations.

LLM Prompt #2

Pose Generation
(Position + Orientation)

Assign a feasible position and orientation to each generated primitive based on the robot, table geometry, primitive properties, and placement constraints.



LLM [1]



Primitives Position and Orientation for **Table 1**, **Table 2**, **Table 3** and **Table 4**

Geometric Validation Collision Checking



✗ Collision Detected

Iterative
Pose Correction
Until Valid



✓ Collision-free Placement